

CITY OF CHEYENNE
CONTRACT MODIFICATION

Approved as to
form only;

Date: 8/2/21

Contract Modification: #2

Project: Happy Jack Landfill Phase 2, Cells 1 and 2 Construction and Cell 3 Excavation
Earthworks (Earthworks base bid)

Project Number: Contract # 7256

Contractor: Dan Hart Patrol Service, LLC.

Date Prepared: July 23, 2021

The above referenced contract is amended or modified as set forth in the following and these amendments and modifications of the plans and specifications are agreed to by the parties. All other aspects of the original agreement remain unchanged and in full force and effect.

I. Description of Changes

Increase Rock Excavation quantities and add costs for the relocation of Duke Energy's maintenance road.

II. New Contract Items

The relocation of Duke Energy's maintenance road.

III. Schedule

No change to contract length.

IV. Summary

Original Contract Amount	\$4,582,929.28
Previous Modifications	\$310,000.00
Contract Amount Prior to this Modification.....	\$4,892,929.28
This Modification	\$1,107,538.00
New Contract Amount.....	\$6,000,467.28

V. Approvals

DATE: _____

City of Cheyenne

Mayor Patrick Collins

/_____
/ATTEST: City Clerk Kristina F. Jones

Recommend approval by City Project Manager / Accepted by Consultant

Solid Waste Professionals of Wyoming, LLC

PLANNING • PERMITTING • DESIGN • QA/QC



DATE: July 22, 2021

TO: Craig LaVoy
Deputy Public Works Director

FROM: Travis Evans, PE

RE: Happy Jack Landfill Rock Excavation and Duke Energy Road Relocation

BACKGROUND

During excavation of the Cells 1, 2 and 3 Dan Hart Patrol (DHP) has encountered considerable amount of unforeseen rock requiring excavation to reach design subgrade elevations. Geotechnical evaluations did identify rock lenses, but based on the data obtained, it appeared the material would be rippable by a large bulldozer. Per the specifications, rock excavation is defined as:

"the event that the rock is encountered of such a hardness and texture that it cannot be excavated with a D-9N Caterpillar tractor with a single shank ripper, or equivalent equipment; or, for a trench excavation, a 235C Caterpillar excavator with a medium stick and a rock ripping bucket, or equivalent equipment, full compensation for all labor, materials, equipment and other incidental costs required to excavate rock to the subgrade elevations shown in the Contract Drawings and/or as required or directed by the CQA MANAGER, maintaining safe excavation and fill slopes, development and maintenance of temporary haul roads, providing traffic control as necessary, and coordination with Site operations."

DHP mobilized a Caterpillar D9 Dozer equipped with a single ripper shank on May 24, 2021. Solid Waste Professionals of Wyoming (SWPW) evaluated several options for the City to consider regarding this matter as described in this memo.

As of July 11, 2021, approximately 15,000 CY of rock excavation has been identified. However, the excavation depths remaining to reach the design subgrade elevations ranges from 10 to 30 feet below current excavation grades. DHP excavated below the current rock excavation area and identified another potential rock excavation area about 8-feet below. To account for unknown future rock excavation quantities, DHP has provided tiered pricing for excavating the rock. DHP's original unit price for rock excavation was \$45. They are willing to reduce their unit price as shown in Table 1.

Table 1
Tiered Rock Pricing

Quantity	Units	Unit Price	Extended Price
15,000	CY	\$39.00	\$585,000.00
20,000	CY	\$35.50	\$710,000.00
25,000	CY	\$31.75	\$793,750.00
30,000	CY	\$29.00	\$870,000.00

ROCK EXCAVATION OPTIONS

Due to the increase cost of rock excavation, there are four options that the City may consider to reduce construction costs as described below:

- **Option 1 – Excavate All Cells to Design Subgrade Elevations**

Option 1 involves continuing excavation on Cells 1, 2 and 3 to design subgrade elevations and amend the contract to include 30,000 CY of rock excavation. It is anticipated that to get to design subgrade elevations approximately 30,000 CY of rock will need to be excavated. The lifetime permit prepared by Golder/SWPW calculates the Phase 2 Net Airspace Volume at 6,196,860 CY. The Variance Application prepared by Golder/SWPW calculates the Site Life Disposal Costs at \$232,438,267 for the life of the Phase 2 landfill. The Life Cycle Cost per CY of Airspace to construct Phase 2 to design elevations is \$37.67/CY.

$$\frac{\$232,438,267}{6,196,860 \text{ CY}} = \$37.67/\text{CY}$$

- **Option 2 – Stop Cell 3 Excavation and Excavate Cells 1 and 2 to Design Grade**

Continue excavation on Cells 1 and 2 to design subgrade elevations and increase the rock excavation quantity to 30,000 CY. Excavation in Cell 3 will not continue except for a 36-foot access ramp which will allow trucks to access Cells 1 and 2. The existing grades will be sloped to grade the haul road and the remaining material will be left in place. Comparing this option to the original design grades, the excavation quantity will be reduced by approximately 296,000 CY and will result in approximately a \$361,120 savings. A figure showing the design change is included as Attachment A.

- **Option 3 – Stop Cell 3 Excavation and Raise Cell 1 and 2 Floor Grades 10-feet.**

Raising Cell 1 and 2 current design floor grades 10-feet will reduce the landfill's capacity by 89,408 CY. The Net Airspace Volume if Cell 1 and 2 design grades were raised 10-feet would reduce the landfill's capacity to 6,080,452 CY.

$$6,196,860 \text{ CY} - 89,408 \text{ CY} = 6,080,452 \text{ CY Net Airspace}$$

The Life Cycle Cost per CY of Airspace to raise Cell 1 and 2 floor grades 10-feet above design elevations is \$38.23/CY compared to \$37.67 per CY for the permitted design.

$$\frac{\$232,438,267}{6,080,452 \text{ CY}} = \$38.23/\text{CY}$$

The lifetime permit estimated average daily volume of loose municipal solid waste (MSW) and construction/demolition (C/D) waste received at the landfill during the first year Phase 2 accepts waste is 359 cy of waste per calendar day. Raising the Cell 1 and 2 floor grades 10-feet will result in decreasing the site life by 0.68 years.

$$89,408 \text{ CY} * \frac{\text{Cal. Day}}{359 \text{ CY}} * \frac{\text{Year}}{365 \text{ Cal. Day}} = 0.68 \text{ Years}$$

- **Option 4 – Stop Cell 3 Excavation and Raise Cell 1 and 2 Floor Grades 20-feet.**
Raising Cell 1 and 2 current design floor grades 20-feet will reduce the landfill's capacity by 198,539 CY. The Net Airspace Volume if Cell 1 and 2 design grades were raised 20-feet would reduce the landfill's capacity to 5,971,321 CY.

$$6,196,860 \text{ CY} - 198,539 \text{ CY} = 5,971,321 \text{ CY Net Airspace}$$

The Life Cycle Cost per CY of Airspace to raise Cell 1 and 2 floor grades 20-feet above design elevations is \$38.93/CY compared to \$37.67 per CY for the permitted design.

$$\frac{\$232,438,267}{5,971,321 \text{ CY}} = \$38.93/\text{CY}$$

The lifetime permit estimated average daily volume of loose municipal solid waste (MSW) and construction/demolition (C/D) waste received at the landfill during the first two years Phase 2 accepts waste is 361 cy of waste per calendar day. Raising the Cell 1 and 2 floor grades 20-feet will result in decreasing the site life by 1.51 years.

$$198,539 \text{ CY} * \frac{\text{Cal. Day}}{361 \text{ CY}} * \frac{\text{Year}}{365 \text{ Cal. Day}} = 1.51 \text{ Years}$$

The four options for rock excavation (assuming 30,000 CY of rock excavation) and construction costs associated with each option are summarized in Table 2. A rock excavation quantity of 30,000 CY was used to provide a conservative estimate of construction costs, but may not be necessary based on subsurface conditions encountered during the remaining excavation activities.

Table 2
Rock Excavation Cost Options

Option	Rock Excavation Price	Soil Excavation Quantity (CY)	Additional Construction Cost	Contract Change
1	\$870,000.00	0	\$0.00	\$870,000.00
2	\$870,000.00	296,000	\$361,120.00	\$508,880.00
3	\$870,000.00	385,408	\$109,077.76	\$399,802.24
4	\$870,000.00	494,539	\$242,217.58	\$266,662.42

While the short-term savings is substantial, the long-term expense associated with reducing airspace is substantial. Table 3 summarizes the value of the airspace lost for each of the options.

Table 3
Airspace Cost

Option	Airspace Lost (CY)	Airspace Value per CY	Airspace Value Lost
1	0	\$37.67	0
2	0	\$37.67	0
3	89,408	\$38.23	\$3,418,067.84
4	198,539	\$38.93	\$7,729,123.27

With the inclusion of the 30,000 CY of rock excavation and the fuel escalation change order, the average unit price for all excavation is \$1.76 per CY. When the project was bid, the median unit price for excavation was \$1.82. Construction costs have risen since the project was bid, and it is highly unlikely the project could be rebid for less than \$1.76 per CY.

DUKE ENERGY ACCESS ROUTE EVALUATION

The construction of the Happy Jack Landfill expansion resulted in the removal of a portion of Duke Energy's road used to access three wind turbines along the west side of the landfill expansion area. SWPW evaluated re-route options provided by Duke; prepared design drawings for the preferred option; obtained construction cost estimates from DHP; and prepared cost estimates for engineering design, surveying, and construction oversight. The Duke Energy Access Route Evaluation is included as Attachment B. In the event the City is required to construct the access road, the estimated cost for design and construction is \$237,538.

Attachments

47A-002-001

REFERENCES

- Golder Associates Inc. (Golder) and Solid Waste Professionals of Wyoming (SWPW). 2018. Lifetime Permit Application for a Solid Waste Operating Permit for the Happy Jack Landfill. January 2018
- Golder and SWPW. 2015. Variance Application for Proposed Happy Jack Landfill Expansion. February 2015

ATTACHMENT A

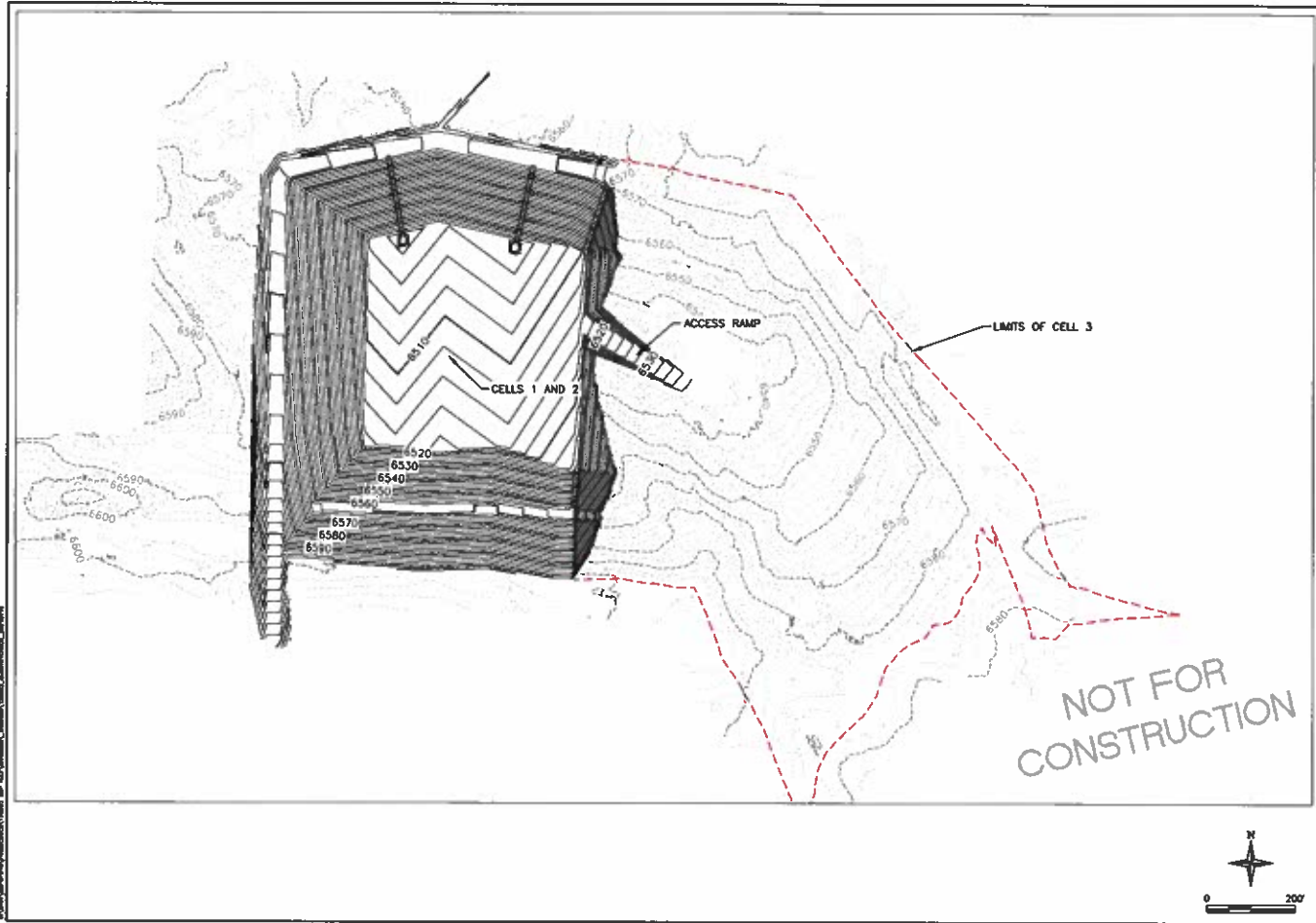


FIG 1	POSSIBLE GRADING MODIFICATION		 Tribble 1825 Commerce Drive Cheyenne, Wyoming 82001 (307) 634-7376 • fax (307) 634-7799	DRAWN BY RS		CHECKED BY MH													
				DATE 7/20/2021		SCALE 1"=40'		FILE		REV		DATE		DESCRIPTION		BY		REVISIONS	
REV																			

ATTACHMENT B

Solid Waste Professionals of Wyoming, LLC

PLANNING • PERMITTING • DESIGN • QA/QC



DATE: July 7, 2021

TO: Craig LaVoy
Deputy Public Works Director

FROM: Travis Evans, PE

RE: Happy Jack Landfill Duke Energy Road Relocation

BACKGROUND

The construction of the Happy Jack Landfill expansion resulted in the removal of a portion of Duke Energy's road used to access three wind turbines along the west side of the landfill expansion area. The City of Cheyenne (City) and Duke Energy (Duke) have discussed alternate locations for an access road over the course of the last several months. On June 1, 2021 the City requested that Solid Waste Professionals of Wyoming (SWPW) evaluate the road alignment options present by Duke. SWPW evaluated the re-route options provided by Duke; prepared design drawings for the preferred option; obtained construction cost estimates from Dan Hart Patrol, Inc. (DHP); and, prepared cost estimates for engineering design, surveying, and construction oversight.

ROUTE EVALUATION

SWPW evaluated the re-route options provided by Duke as shown in Attachment A and re-route options provided by the City in Attachment B.

The reroute option provide by Duke is not feasible, as the route is in the same location as the spoil stockpiles constructed as part of the landfill expansion project. Additionally, Duke suggested salvaging the roadway surfacing from the abandoned access road along the south side of the landfill expansion. A large portion of this road has been removed as part of the landfill project and salvaging the remaining road base will be challenging due to the woven geotextile placed beneath the road base. SWPW recommends installing new road base over the new access road.

The City provided two route options. Option 1 is approximately 3,800 LF and is located along the north boundary of the new spoil Stockpile 2 and Stockpile 3. Option 2 utilizes the existing landfill entrance and is approximately 533 LF long. SWPW determined Option 2 was not feasible due to the overhead power lines and power poles which would need to be relocated.

The route SWPW ultimately chose for the access road is similar to the route identified in Option 1 presented by the City. The route can be found in the Construction Drawings provided as Attachment C.

ROAD DESIGN

The road was designed to incorporate the General Site Requirements (ATS, Date Unknown) provided as Attachment D; and, Happy Jack Road As-Build (sic) Drawings provided via email (Vereschagin 2021). The following design elements were incorporated into the road design:

- The roadway is 16-feet wide and 23-feet wide through 90-degree horizontal curves
- The inside edge of travel way through 90-degree horizontal curves has a minimum radius of 135-feet
- Vertical curves will not exceed 6-inches of elevation difference over 50-feet
- Topsoil will be stripped to a depth of 4-inches and will be replaced after completion of roadway construction. Topsoil will be blended to match the gravel shoulders and graded to match existing native ground elevations and promote drainage
- The roadway design matches existing ground elevations as closely as possible while meeting the minimum design requirements to provide sheet flow across the roadway
- Native subgrade will be moisture conditioned and compacted to a thickness of one foot. The specified density and moisture content is 95% of the maximum dry density and plus or minus 3% of the optimum moisture content per ASTM D698. The subgrade will be proof-rolled prior to placing the geotextile fabric
- A 10-ounce woven geotextile fabric will be placed beneath the gravel surfacing
- The gravel surfacing is 8-inches thick. The specified density and moisture content is 98% of the maximum dry density and plus or minus 2% of the optimum moisture content per ASTM D698

The construction specifications are provided in the Construction Drawings provided as Attachment C.

ESTIMATED CONSTRUCTION COST

SWPW provided the design drawings and estimated quantities to DHP. Steven Gerber with DHP submitted a cost estimate via email on July 6, 2021. DHP's cost estimate is summarized in Table 1 below:

Table 1
Access Road Construction
Construction Cost Estimate

Item	Description	Estimated Quantity	Units	Unit Price	Extended Price
1.	Embankment	500	CY	\$5.00	\$2,500
2.	Unclassified Excavation	1,500	CY	\$5.00	\$7,500
3.	Gravel Surfacing	2,600	CY	\$44.00	\$114,400
4.	Topsoil Strip and Replace (4")	1,300	CY	\$5.60	\$7,280
5.	Woven Geotextile	106,000	SF	\$0.30	\$31,800
6.	Subgrade Preparation	11,800	SY	\$2.29	\$27,022
7.	Reclamation	3.5	ACRE	\$5,500	\$19,250
Total Estimated Construction Cost					\$209,752

SWPW's detailed cost estimate for design, construction surveying, and construction oversight is provided as Attachment D. SWPW's estimated fees to support road construction is \$27,786.

Total engineering and construction costs are estimated to be \$237,538.

Attachments

47A-002-001

REFERENCES

ATS Wind Energy Services. Unknown. General Site Requirements for Suzlon Wind Energy. St. Cloud: ATS Wind Energy Services. 4-12 p.

Vereschagin, S. 2021. RE: Happy Jack Road expansion. June 23 [Online]. Available e-mail: Sasha.Vereschagin@duke-energy.com

ATTACHMENT A



ATTACHMENT B



Janice L. Walker
Associate General Counsel
139 East Fourth Street
M/C 1212-M
Cincinnati, Ohio 45202

o: 513.287.4362

c: 513.477.7279

f: 513-287-4386

E-mail: Janice.walker@duke-energy.com

July 1, 2021

VIA EMAIL modonnell@cheyennecity.org

City Attorney's Office
Mr. Michael R. O'Donnell
City Attorney
Municipal Building
2101 O'Neil Avenue, Room 308
Cheyenne, Wyoming 82001

Dear Mr. O'Donnell:

Thank you for your response. It is unfortunate that the City and Happy Jack are unable to reach an amicable resolution to this matter.

Your letter concedes that the City took a portion of the road paid for and built by Happy Jack for the City's own purpose to operate its own landfill. The City's actions constitute inverse condemnation. See *Cheyenne Airport Bd. v. Rogers*, 707 P.2d 717, 730 (Wyo. 1985).

Accordingly, Happy Jack is entitled to compensation for the value of the road taken, and would be entitled to pursue its fees and costs should litigation be commenced. The City — as an entity with condemnation authority—cannot simply destroy and take Happy Jack's road without providing just compensation. The lease does not give the City the right to destroy or remove improvements constructed by Happy Jack.

We expect the City to provide just compensation for the taking of Happy Jack's road, including the full value of the road taken.

Please confirm the City will provide such compensation. If necessary, Happy Jack can hire an appraiser to value the interest and road taken, which undoubtedly will be more expensive than simply constructing a new road, but it may be more cost effective for everyone to continue discussions involving a solution whereby the City agrees to pay for the road it has taken.

Sincerely,

Janice L. Walker

Janice L. Walker

cc: Vicki Nemecek, Director of Public Works, City of Cheyenne (via email)
Craig LaVoy, Deputy Director of Public Works, City of Cheyenne (via email)
Sasha Vereschagin, Duke Energy Mgr., Comm. Renewable Ops (via email)
Benjamin Jordan, Duke Energy Director, Comm. Renewable Ops (via email)



Janice L. Walker
Associate General Counsel

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May 26, 2021

VIA EMAIL modonnell@cheyennecity.org AND CERTIFIED MAIL

Michael R. O'Donnell
City Attorney's Office
Municipal Building
2101 O'Neil Avenue, Room 308
Cheyenne, Wyoming 82001

RE: HAPPY JACK WIND FARM, CHEYENNE WYOMING ACCESS ROAD ISSUE

Dear Mr. O'Donnell:

Happy Jack Wind (HJW) is in receipt of your letter dated May 14, 2021 in response to HJW's initial letter dated May 11, 2021, and we understand the City's position; however, based upon our interpretation of the Easement, we respectfully disagree. We have a good working relationship with the City and we certainly endeavor to continue to foster that relationship in the midst of a disagreement on whether the City or HJW is responsible for the cost and expense of a replacement access road (HJW Access Road) that HJW constructed at its cost per the Easement and the City removed as a part of the construction of its landfill.

First, a point of clarification: access to the HJW turbines is a part of the operation and maintenance of the HJW wind energy system. Second, the previous HJW Access Road that was removed by the City. This HJW Access Road was part of the HJW wind energy project, a road that did not previously exist (prior to the Easement), and a road which HJW constructed at its sole cost and expense. During the planning and construction of the City's landfill, the City removed the road subsequent to the City's operations representatives communicating to the HJW operational team that the City would construct a replacement access road subsequent to the completion of the City's landfill project; however, when the time came, the City renege on its previous agreement to construct a new access road.

HJW paid to construct a road that had not previously existed, i.e., the road was not a pre-existing City road for which the City paid for and consented for HJW to use. Thus, it is HJW's position that the City is responsible for the expense to replace the previous HJW Access Road with a similarly situated road in that the removal of the HJW Access Road was inconsistent with HJW's use of the its Access Road pursuant to the Easement, i.e., it was completely removed. In summary, the parties simply disagree on who is responsible for the expense of a replacement road, not HJW's ability to construct a replacement access road. HJW reserves the right to resolve this issue by the appropriate legal means at some future point.

Notwithstanding the above, HJW would like to proceed expeditiously with the construction of a replacement access road. As such, we have requested estimates on the cost, looking at permitting and engineering, and the other requisite requirements and approvals for HJW to construct the

replacement road. Thus far we have determined that for HJW or its designee to gear up to construct the road, timing is a significant issue. Therefore, since the City currently has the infrastructure in place to construct the road, and as such, the City's construction of a replacement access road may be a more expeditious and economical approach to completing an access road.

Based upon the immediately preceding paragraph, we are requesting that the City provide estimates for Options 1, 2 and 3 below if the City were to construct a replacement access road from the three options below using the attached depiction of the area:

- **OPTION 1:**
- Preferred route:
- T14 Route:
 - New road to spec between existing Duke Energy site entrance to T14
 - Shown in green in snip attached

- **OPTION 2:**
- T12 & T13 Route
 - Repair road to spec between T12 and T13
 - Shown in blue in snip attached

- **OPTION 3:**
- Alternate route option:
- Landfill entrance route:
- New road with gate and road tubes for drainage between Landfill entrance to T14.
 - Shown in red in attached snip
 - Gate and road tube area shown yellow in attached snip

Time is of the essence and we would ask the City to provide the three estimates for constructing each of the three options by Friday June 4th or as early as you can provide such information. This would be very helpful for HJW's overall determination in the appropriate path forward.

Thank you for your time and consideration.

Sincerely,

Janice L. Walker

Janice L. Walker

Attachment: Drawing of Area Depicting Options 1-3 above.

Copies to: Vicki Nemecek, Director of Public Works, City of Cheyenne (via email)
Craig LaVoy, Deputy Director of Public Works, City of Cheyenne (via email)
Ben Jordan, Director Renewable Operations, Happy Jack Wind (via email)
Sasha Vereschagin, Manager Renewable Operations, Happy Jack Wind (via email)



ATTACHMENT C

PROJECT #47A-002-001

[M	Rip Sheet
1	General Notes
2	Overview
PP.1 - PP.2	Plan and Profile Sheets



PROJECT LOCATION

NOT FOR
CONSTRUCTION
REVIEW ONLY

A	INVEST	NAME FOR SLIP REVIEW	NO	NO	NO	NO
REV	DATE	REASON FOR DISAPPROVAL	DISCARD	CARD	DISCARD	REVIEW
PROJECT HAPPY JACK WIND ACCESS ROAD DUKE ENERGY CHEYENNE, WY						
		DATE REC'D TITLE SHEET FILE NO. <i>44-1987-Subj</i> PROJECT NO. <i>44-1987-101</i> T-01				

A. GENERAL

1. ALL QUANTITIES ARE CONSIDERED APPROXIMATE AND ARE FOR ESTIMATING PURPOSES ONLY.
2. THE CONTRACTOR SHALL INCLUDE ALL MATERIALS, TOOLS, EQUIPMENT, LABOR, AND APPROPRIATE PRICES TO COMPLETE THE WORK WITHIN THE BID PRICE.
3. THE CONTRACTOR SHALL BECOME FAMILIAR WITH THE SITE PRIOR TO BIDDING AND CONSTRUCTION.
4. LOCAL LOAD LIMIT REQUIREMENTS SHALL BE ENFORCED ON ALL STATE HIGHWAYS, CITY STREETS, AND COUNTY ROADS.

B. EXISTING UTILITIES

1. CONTRACTOR SHALL NOTIFY ALL UTILITY COMPANIES WITH A 48 HOUR ADVANCED NOTICE AND DETERMINE THE EXACT LOCATION OF ALL UNDERGROUND UTILITIES BEFORE COMMENCING WORK. CONTRACTOR IS RESPONSIBLE FOR PROTECTING AND PROPERLY REPAIRING DAMAGED UTILITIES.
2. UTILITY LOCATIONS SHOWN ON THE DRAWINGS ARE APPROXIMATE BASED ON RECORD OF UTILITY COMPANIES AND FIELD VERIFICATION BY THE UTILITY COMPANIES. FIELD VERIFICATION OF BURIED GAS, ELECTRIC, TELEPHONE, WATER, SEWER, AND CABLE TV LINES ARE BY ELECTRONIC OR MAGNETIC DETECTION METHODS. ALL UTILITY LOCATIONS ARE SUBJECT TO THE ACCURACY OF THE LOCATION METHOD, AND SUBJECT TO RELOCATION FROM THE TIME THAT THE DRAWINGS WERE PREPARED. NO EXCAVATION SHALL BE PERFORMED.
3. THE CONTRACTOR SHALL NOTIFY THE OWNER OR ENGINEER OF ALL UTILITIES ENCOUNTERED DURING CONSTRUCTION AND SHALL NOT BURN OR LIFT. THE CONTRACTOR HAS MADE A RECORD OF ITS TYPE, SIZE, AND LOCATION.
4. CONTRACTOR SHALL CONTACT ONE CALL OF WYOMING FOR UTILITY LOCATES AT 811.
5. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROTECT, ADJUST, AND RELOCATE ANY UTILITIES INTERFERING WITH THE SPECIFIED CROSS-SECTIONS AND DEPTHS AS DESIGNED IN THE SITE PLAN AND GRADING PLANS.

C. CONSTRUCTION LIMITS

1. CONTRACTOR SHALL RESPECT ALL RIGHT-OF-WAY, EASEMENT, AND CONSTRUCTION PERMIT BOUNDARIES. ALL WORK SHALL BE DONE WITHIN THESE BOUNDARIES. ANY WORK OUTSIDE OF THE RIGHT-OF-WAY, EASEMENTS, OR CONSTRUCTION PERMIT SHALL ONLY BE DONE AFTER RECEIVING WRITTEN PERMISSION FROM THE PROPERTY OWNER. THIS PERMISSION SHALL BE OBTAINED BY THE CONTRACTOR. A COPY OF THE WRITTEN PERMISSION SHALL BE PROVIDED TO THE OWNER AND ENGINEER PRIOR TO WORK OUTSIDE OF THE ESTABLISHED BOUNDARIES. CONTRACTOR IS FULLY LIABLE FOR THE COST OF ANY DAMAGES TO PROPERTY AS WELL AS THE SPENT BY THE OWNER AND ENGINEER AT THEIR STANDARD HOURLY RATES TO REMEDY SUCH CIRCUMSTANCES WITH THE PROPERTY OWNER.

D. SAFETY

1. GENERAL. THE CONTRACTOR IS SOLELY RESPONSIBLE FOR DEMONSTRATING AND CONSTRUCTING STABLE, TEMPORARY EXCAVATIONS AND SHOULD SHORE, BLOKE, OR BRACE THE SIDES OF THE EXCAVATIONS AS REQUIRED TO MAINTAIN STABILITY OF BOTH THE EXCAVATION SIDES AND BOTTOM. ALL EXCAVATIONS SHOULD COMPLY WITH APPLICABLE LOCAL, STATE, AND FEDERAL REGULATIONS INCLUDING THE CURRENT OSHA EXCAVATION AND TRENCH SAFETY STANDARDS. CONSTRUCTION SITE SAFETY IS THE SOLE RESPONSIBILITY OF THE CONTRACTOR. WHO SHALL ALSO BE SOLELY RESPONSIBLE FOR THE MEANS, METHODS, AND SEQUENCING OF CONSTRUCTION OPERATIONS. UNDER NO CIRCUMSTANCES SHOULD THE INFORMATION PROVIDED BE INTERPRETED TO MEAN THAT THE ENGINEER IS ASSUMING RESPONSIBILITY FOR CONSTRUCTION SITE SAFETY OR THE CONTRACTOR'S ACTIVITIES. SUCH RESPONSIBILITY IS NOT BEING IMPARTED AND SHALL NOT BE IMPERED.
2. EXCAVATION AND BLOKE. IN NO CASE SHOULD BLOKE HEIGHT, BLOKE INCLINATION, OR EXCAVATION DEPTH EXCEED UTILITY TRENCH EXCAVATION DEPTH. EXCEED THOSE SPECIFIED IN LOCAL, STATE, AND FEDERAL SAFETY REGULATIONS. ESPECIALLY, THE CURRENT OSHA HEALTH AND SAFETY STANDARDS FOR EXCAVATIONS. 29 CFR PART 1926 SHOULD BE FOLLOWED. IT IS THE ENGINEER'S UNDERSTANDING THAT THESE RECOMMENDATIONS ARE BEING STRICTLY ENFORCED AND IF THEY ARE NOT CLOSELY FOLLOWED, THE CONTRACTOR COULD BE LIABLE FOR PUNITIVE DAMAGES.
3. CONTRACTOR SHALL CLEARLY MARK AND/OR FENCE ALL OBSTRUCTIONS, EXCAVATIONS, AND CONSTRUCTION MATERIALS AND EQUIPMENT. THE PROJECT SUPERINTENDENT FOR A DESIGNATED REPRESENTATIVE SHALL MONITOR THE PROJECT FOR SAFETY CONCERNS AND POTENTIAL HAZARDS AT ALL TIMES.

E. RECLAMATION

1. THE CONTRACTOR SHALL RECLAIM ALL AREAS OF DISTURBANCE TO A CONDITION THAT IS EQUAL TO OR BETTER THAN THE ORIGINAL INCLUDING TOPSOIL. THE CONTRACTOR SHALL SALVAGE AND STOCKPILE UNPAVED TOPSOIL. THE CONTRACTOR SHALL RECLAIM ALL DISTURBED AREAS.

F. EROSION

1. CONTRACTOR SHALL OBTAIN AND IMPLEMENT SWPPP PRIOR TO CONSTRUCTION. COSTS ASSOCIATED WITH IMPLEMENTATION OF SWPPP SHALL BE LUMP SUM. CONTRACTOR SHALL MAINTAIN THE SWPPP THROUGHOUT THE PROJECT AND REQUIRED THEREAFTER OF THE SWPPP.

GENERAL NOTES:

G. SURVEY

1. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ALL CONSTRUCTION SURVEY AND LAYOUT.
2. SURFACES AND CONTROL SHALL BE SUPPLIED TO THE SUCCESSFUL CONTRACTOR FOLLOWING AWARD OF CONTRACT.

H. SUBMITTALS

1. CONTRACTOR SHALL SUBMIT ONE ELECTRONIC COPY OF THE SPECIFIED SUBMITTALS TO THE ENGINEER FOR REVIEW.
2. ENGINEER SHALL REVIEW ALL SUBMITTALS FOR CONFORMANCE WITH THE DRAWINGS AND SPECIFICATIONS.
3. CONTRACTOR IS RESPONSIBLE FOR CORRECTING DEFECTIVE CONSTRUCTION REGARDLESS OF REVIEW RESULT.
4. CONTRACTOR SHALL SUBMIT A SUBGRADE SOIL GRAIN SIZE ANALYSIS, MOISTURE CONTENT, AND STANDARD PROCTOR MAXIMUM DRY DENSITY TEST FOR EVERY 1500 LINEAR FEET OF ROADWAY INSTALLED.
5. CONTRACTOR SHALL SUBMIT A ROAD BASE GRAIN SIZE ANALYSIS, MOISTURE CONTENT, AND STANDARD PROCTOR MAXIMUM DRY DENSITY TEST FOR EVERY 1500 LINEAR FEET OF ROADWAY INSTALLED.

I. PRODUCTS

1. ROAD BASE SHALL BE WYDOT CRUSHED BASE (GRAVEL ROAD) TYPE "W", MEETING THE REQUIREMENTS OF SECTION 604.4.
2. ROAD SURFACING SHALL BE COMPACTED NATIVE SOIL WITH VEGETATION CLASSIFICATION OF CL, CLAL, OR SC-BL.
3. RECLAMATION NEED SHALL MEET THE REQUIREMENTS OF THE LANDFILL PROJECT MANUAL.
4. WATER FOR DUST CONTROL AND MOISTURE DENSITY CONTROL OF EMBANKMENT AND GRAVEL SURFACING TO BE PROVIDED BY THE CONTRACTOR AND IS INCIDENTAL TO APPLICABLE BID ITEMS.

J. EROSION

1. REMOVE TOPSOIL AND ORGANICS FROM THE WORK AREA AND STOCKPILE FOR REPLACEMENT AFTER ROADWAY CONSTRUCTION. STRIPPING DEPTHS WILL VARY AND SHOULD BE EVALUATED PRIOR TO CONSTRUCTION.
2. THE ENTIRE ROADWAY LENGTH AND WIDTH UNDER THE GRAVEL SURFACING SHALL BE MOISTURE CONDITIONED AND COMPACTED TO A DEPTH OF 12 INCHES.
3. PRIOR TO PLACEMENT OF GEOTEXTILE, ENGINEER SHALL OBSERVE PROFILES OF THE ENTIRE ROADWAY LENGTH TO IDENTIFY UNSTABLE SUBGRADE, PROFILES SHALL BE A FULLY LOADED TANDER AXLE DUMP TRUCK WITH MINIMUM GROSS WEIGHT OF 20 TONS. NO RUTTING GREATER THAN 1.5 INCHES OR PUMPING SHOULD BE OBSERVED BEHIND THE LOADED TRUCK.
4. ROAD BASE SHALL BE TO MINIMUM THICKNESS SPECIFIED AND COMPACTED TO A MINIMUM 98% STANDARD PROCTOR AT 4% OF OPTIMUM MOISTURE CONTENT. CROWN AND CROSS GRADE ON THE ROAD SHOULD ALLOW FOR FUTURE DRAINAGE.
5. CONTRACTOR IS RESPONSIBLE FOR ANY CLEANING AND GRUBBING NECESSARY DURING THE REMOVAL OF TOPSOIL.
6. RECLAMATION SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF THE LANDFILL PROJECT MANUAL.

K. TESTING AND INSPECTION

1. SUBMIT ONE COPY OF ALL INSPECTION AND TESTING RECORDS TO THE ENGINEER.
2. ENGINEER SHALL REVIEW ALL INSPECTION AND TESTING RECORDS FOR CONFORMANCE TO THE DRAWINGS AND SPECIFICATIONS. THIS REVIEW DOES NOT RELIEVE THE CONTRACTOR FROM CORRECTING DEFECTIVE WORK.
3. COPIES OF ALL INSPECTIONS AND TESTING RESULTS SHALL BE KEPT ON SITE FOR OWNER REVIEW.
4. TESTING FREQUENCIES ARE AS FOLLOWS:
 - A. SUBGRADE SOILS (FOR EACH TYPE ENCOUNTERED)
 - 1 GRAIN SIZE ANALYSIS
 - 1 NATURAL MOISTURE CONTENT
 - 1 STANDARD PROCTOR MAXIMUM DRY DENSITY TEST (EMPHASIS)
 - B. ROAD BASE (FOR EVERY 1500 LF OF ROAD)
 - 1 GRAIN SIZE ANALYSIS
 - 1 NATURAL MOISTURE CONTENT
 - 1 STANDARD PROCTOR MAXIMUM DRY DENSITY TEST

L. EMBANKMENT

1. EMBANKMENT SHALL BE BY CUBIC YARD (CY) BASED ON A FIELD SURVEY OF THE MATERIAL PLACED BY THE ENGINEER. ITEM SHALL INCLUDE ALL LABOR, MATERIAL, EQUIPMENT, AND INCIDENTALS REQUIRED TO EXCAVATE, TRANSPORT, PLACE, AND COMPACT EMBANKMENT MATERIAL WITHIN THE BOUNDARIES OF THE ROADWAY TO THE ELEVATIONS SHOWN ON THE DRAWINGS. ESTIMATED QUANTITIES ARE APPROXIMATE IN-PLACE QUANTITIES WITH SHrink FACTORS AS SHOWN AND ARE TO BE USED ONLY AS A BASIS FOR ESTIMATING THE PROBABLE COST OF THE FILL EXCAVATED, PLACED, AND STOCKPILED.
2. UNCLASSIFIED EXCAVATION SHALL BE BY CUBIC YARD (CY) BASED ON A FIELD SURVEY OF THE MATERIAL PLACED BY THE ENGINEER. ITEM SHALL INCLUDE ALL LABOR, MATERIAL, EQUIPMENT, AND INCIDENTALS REQUIRED TO EXCAVATE, TRANSPORT, AND STOCKPILE UNCLASSIFIED MATERIAL TO THE ELEVATIONS SHOWN ON THE DRAWINGS. ESTIMATED QUANTITIES ARE APPROXIMATE IN-PLACE QUANTITIES AND ARE TO BE USED ONLY AS A BASIS FOR ESTIMATING THE PROBABLE COST OF THE FILL EXCAVATED, PLACED, AND STOCKPILED.
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L. MEASUREMENT & PAYMENT

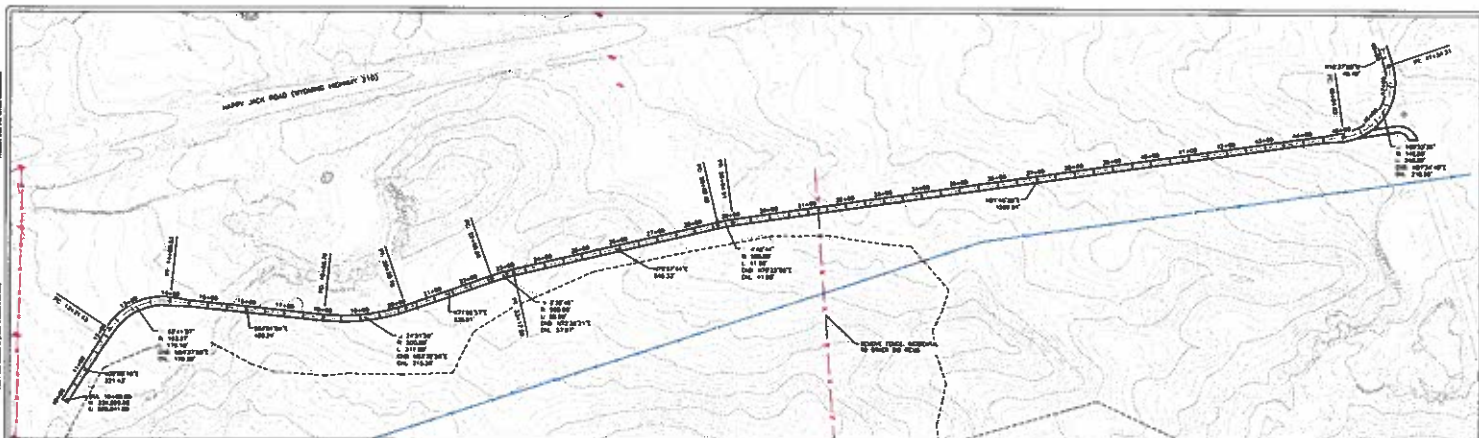
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EXPLANATION

SYMBOL	DESCRIPTION
	POWER POLE
	10' x 10' x 10' CONCRETE
	12' x 12' x 12' CONCRETE
	18' x 18' x 18' CONCRETE
	24' x 24' x 24' CONCRETE
	30' x 30' x 30' CONCRETE
	36' x 36' x 36' CONCRETE
	42' x 42' x 42' CONCRETE
	48' x 48' x 48' CONCRETE
	54' x 54' x 54' CONCRETE
	60' x 60' x 60' CONCRETE
	66' x 66' x 66' CONCRETE
	72' x 72' x 72' CONCRETE
	78' x 78' x 78' CONCRETE
	84' x 84' x 84' CONCRETE
	90' x 90' x 90' CONCRETE
	96' x 96' x 96' CONCRETE
	102' x 102' x 102' CONCRETE
	108' x 108' x 108' CONCRETE
	114' x 114' x 114' CONCRETE
	120' x 120' x 120' CONCRETE
	126' x 126' x 126' CONCRETE
	132' x 132' x 132' CONCRETE
	138' x 138' x 138' CONCRETE
	144' x 144' x 144' CONCRETE
	150' x 150' x 150' CONCRETE
	156' x 156' x 156' CONCRETE
	162' x 162' x 162' CONCRETE
	168' x 168' x 168' CONCRETE
	174' x 174' x 174' CONCRETE
	180' x 180' x 180' CONCRETE
	186' x 186' x 186' CONCRETE
	192' x 192' x 192' CONCRETE
	198' x 198' x 198' CONCRETE
	204' x 204' x 204' CONCRETE
	210' x 210' x 210' CONCRETE
	216' x 216' x 216' CONCRETE
	222' x 222' x 222' CONCRETE
	228' x 228' x 228' CONCRETE
	234' x 234' x 234' CONCRETE
	240' x 240' x 240' CONCRETE
	246' x 246' x 246' CONCRETE
	252' x 252' x 252' CONCRETE
	258' x 258' x 258' CONCRETE
	264' x 264' x 264' CONCRETE
	270' x 270' x 270' CONCRETE
	276' x 276' x 276' CONCRETE
	282' x 282' x 282' CONCRETE
	288' x 288' x 288' CONCRETE
	294' x 294' x 294' CONCRETE
	300' x 300' x 300' CONCRETE

NOT FOR
CONSTRUCTION

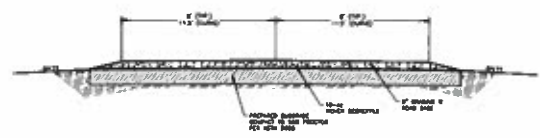
DATE FOR REVIEW		NO	NO	NO	NO
BY	DATE	REVIEW	REVISION	DATE	REVISION
HAPPY JACK WIND ACCESS ROAD DUKE ENERGY CHEYENNE, WY					
DESIGNER					
GENERAL NOTES					
FILE NAME: nrls_2024_0000.dwg					
PLOT NO. 1					



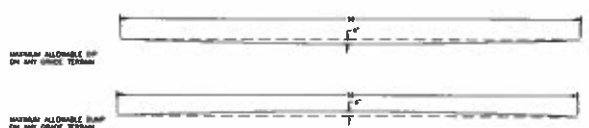
ESTIMATED QUANTITIES

ITEM	QTY
Gravel Road	100,000 CY
Gravel Sub-base	100,000 CY
Gravel Base	100,000 CY
Gravel Surface	100,000 CY
Gravel Shoulder	100,000 CY
Gravel Easement	100,000 CY
Gravel Right-of-Way	100,000 CY

Notes:
1. All quantities are based on the proposed plan and profile.



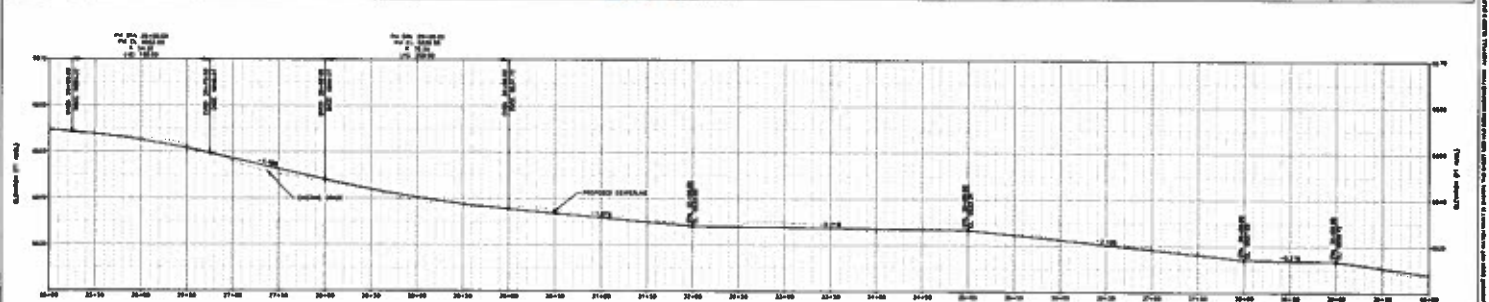
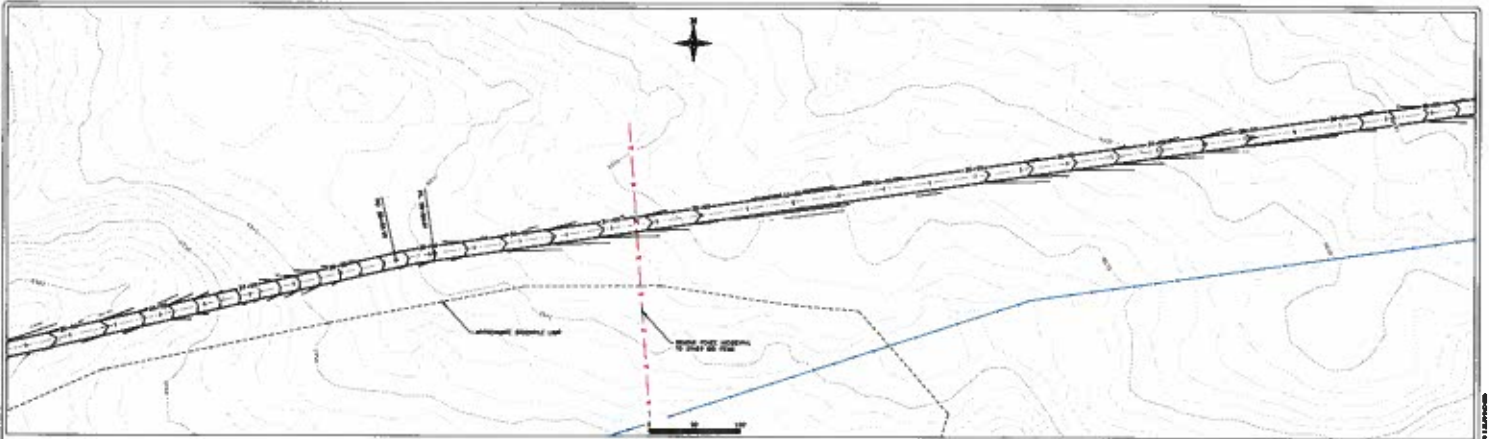
Ⓐ TYPICAL ROADWAY SECTION



Ⓑ TYPICAL ROADWAY SECTION

NOT FOR
CONSTRUCTION

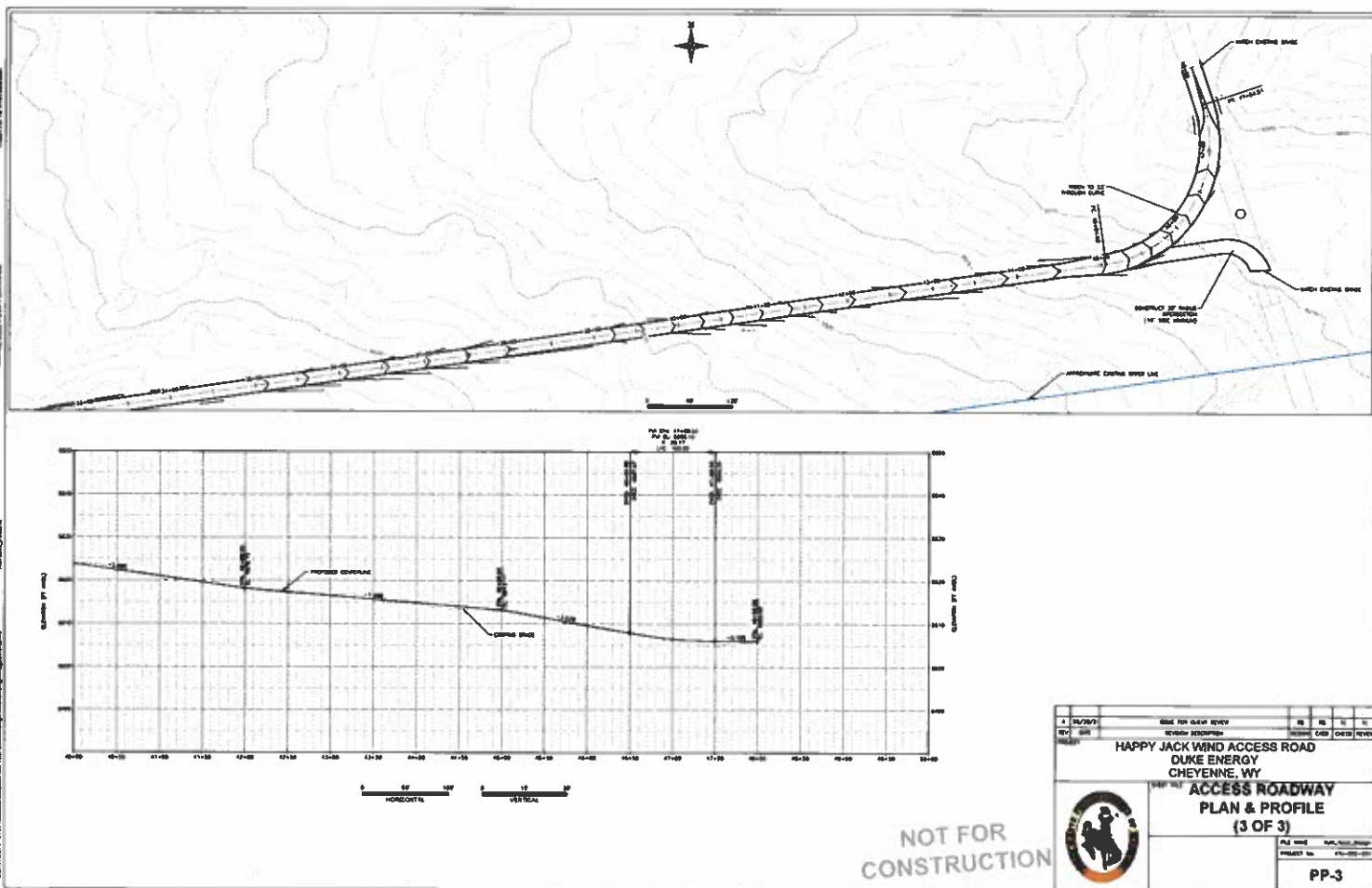
DATE FOR REVIEW		BY	RE	TE	RE
REVIEW DESCRIPTION		REVIEW	DATE	REVIEW	REVIEW
HAPPY JACK WIND ACCESS ROAD DUKE ENERGY CHEYENNE, WY					
ACCESS ROADWAY OVERVIEW					
PROJECT NO.		2			



0 50 100 HORIZONTAL
0 10 20 VERTICAL

NOT FOR
CONSTRUCTION

HAPPY JACK WIND ACCESS ROAD DUKE ENERGY CHEYENNE WY		DATE: 10/1/10		DRAWN: J. L. HARRIS	
ACCESS ROADWAY PLAN & PROFILE (2 OF 3)		SCALE: 1" = 40'		SHEET: 2 OF 3	
PP-2		APP. NAME: J. L. HARRIS		APP. NO: 1000000000	



ATTACHMENT D



ATS Wind Energy Services General Site Requirements

**For
Suzlon Wind Energy**

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Section 7. Turn-a-rounds	13
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Basic Requirements

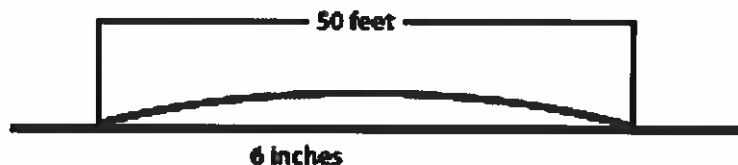
1. All public roads (County, township, local, etc.) that are to be used to access the wind farm shall be able to safely support the movement of the oversize, overweight loads in all weather conditions.
2. All site access roads shall be able to safely support the movement of the oversize, overweight loads. In all weather condition
3. Trucks shall be able to safely enter, unload and exit the wind farm. In all weather conditions.
4. All roads shall have in place areas designated as passing areas. In order to allow a smooth entrance and egress from the site. Allowing loaded trucks, cars and other equipment to pass without danger to personal or equipment.

Road Requirements

Road width can be critical. The roads (in a straight line) must be no less than 16' wide with proper compaction all the way to outside of the 16'. With uneven site roads, trailers will not always track in a perfect straight line and must have a wider base to travel in.

Road crown (hump) 12" within 100' and 6" in 50' are the allowable amounts for the tower trucks in the raised position as well as Nacelle trucks. The 12" crown (hump) needs to be spread out over the 100', to prevent damage to loads or equipment.

Example of Maximum Allowable Hump on Flat Terrain

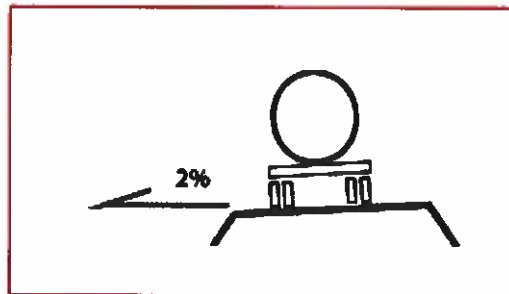
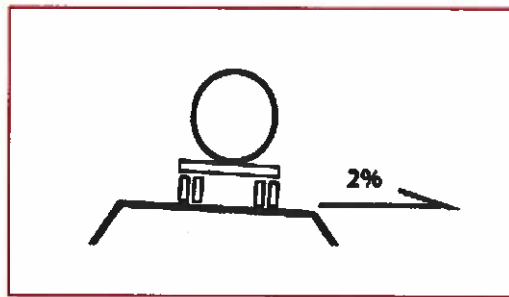


Maximum Slope Terrain 6%



Road Requirements (continued)

Road Side-Slope shall be as level side to side as possible still allowing for water run off. 2% shall be the maximum bank allowed to prevent roll over of top-heavy loads and to prevent trailers from sliding sideways during inclement weather conditions.



Maximum Bank 2%

Road Surface shall be of sufficient compaction to support the weight of loaded truck in all weather conditions. The Nacelle has a gross weight of 260,000lbs. and an axle load of 20,000lbs. per axle. (See load specifications)

Turns

All turns on public roads directly leading to the site and all site access/string roads that loaded trucks must travel on shall follow the same requirements.

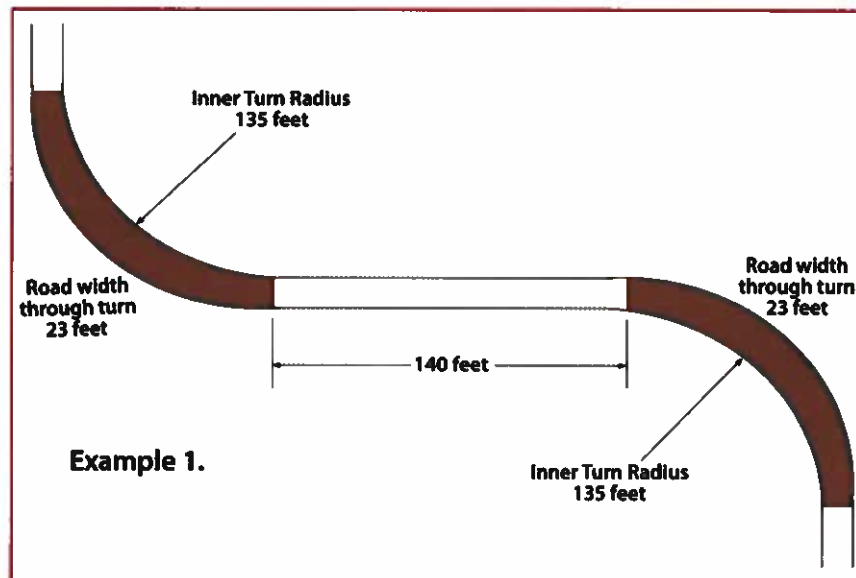
All turns shall have an inner radius of 135FT.

Turns shall not have banks of more than 2% through the duration of the turn.

Turns shall be have no more than a 4% up or down grade through the duration of the turn. Should a turn have to be built on a hill with more than a 4% grade the maximum curve will be 30-degrees and the road width will need to be increased. The extra width will be needed when an assist vehicle is used.

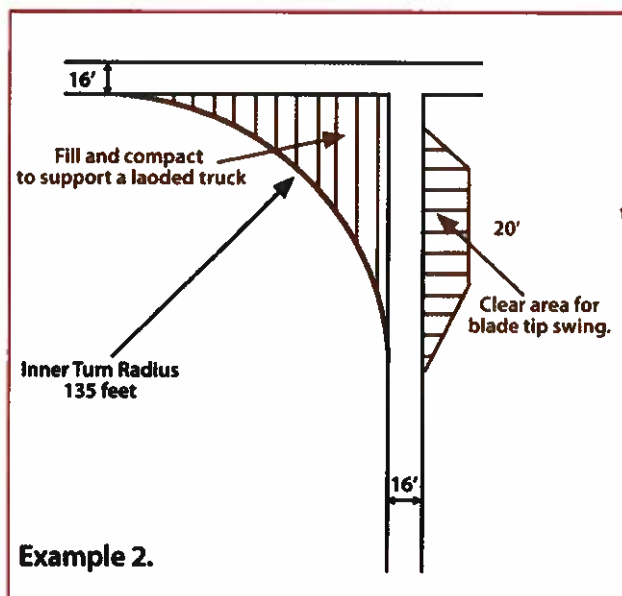
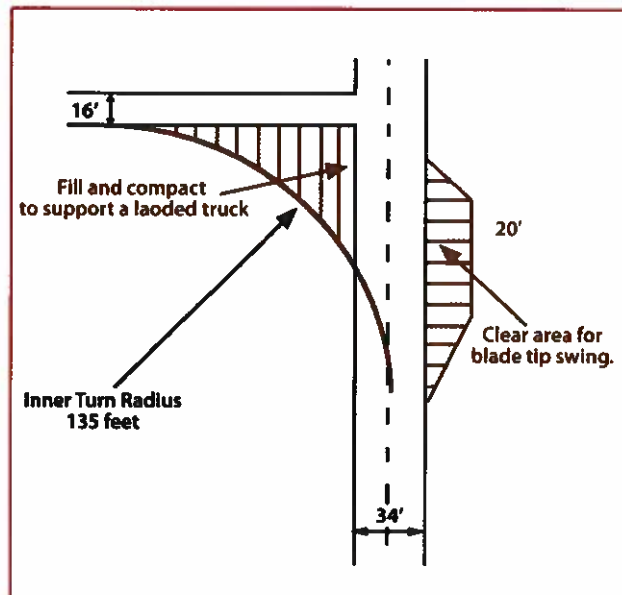
This can be addressed on case-by-case basis.

Turns (continued)



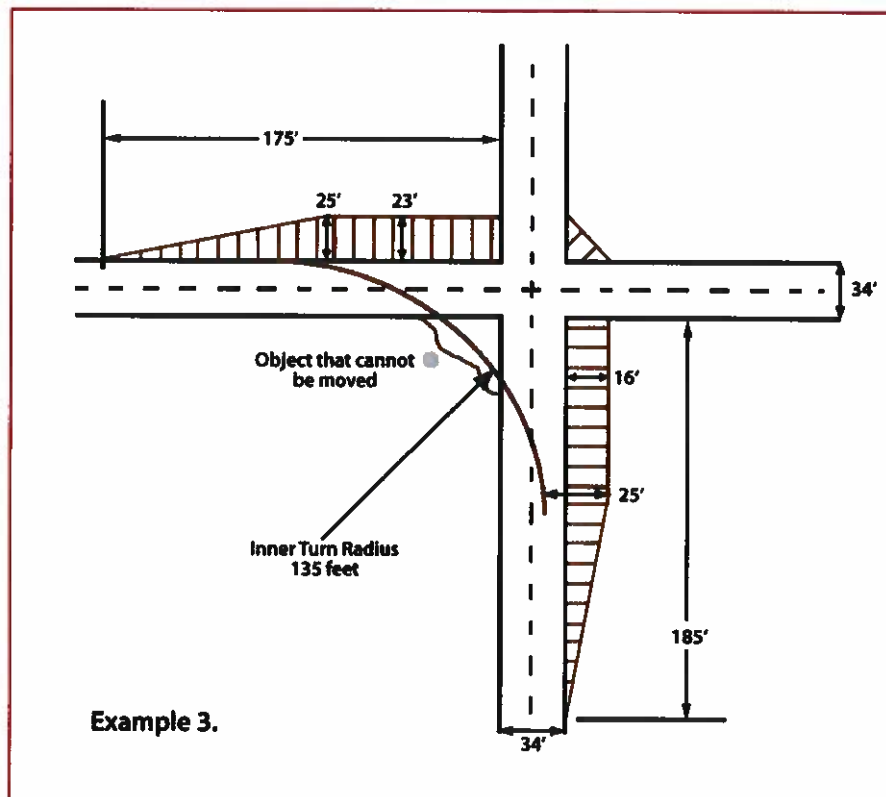
Turn Example 1. Any 90-degree turn on a job site must have a 135' inner radius and widen to 23' wide for the duration of the turn. It must have a straight area of at least 140' before the next turn. This will give the trailer enough time to track back in line before entering the next turn. There will need to be adequate cleared area on the outside of the curve to allow for Blade tip swing.

Turns (continued)



Turn Example 2. 90-degree turns: from a 16-foot wide road to 16-foot wide road and multi lane to 16-foot wide road. Showing cleared area for the swing of the blade tip.

Turns (continued)



Example 3: a turn with a fixed object that cannot be removed.

If an object can not be removed from the inside of a corner fill may be added to one or more of the other sides of the intersection in order to achieve the needed 135FT inner radius.

This example shows one way of obtaining the needed radius.

Curves

All curves shall have the same inner radius of 135FT. The curve shall widen from a 16FT wide road to a 23FT wide road for the duration of the curve to allow for the off tracking of the trailer.

Curves shall have no more than a 2% bank to prevent trailers from sliding off the road during inclement weather.

Curves shall not have an up or down grade of more than 4%. If the curve has a grade of more than 4% then the road width may need to be increased to allow for the added off track of the trailer when an assist vehicle is used.

This can be looked at on case-by-case bases.

Curves shall be separated by a 140 FT long stretch of straight road before curving back in the opposite direction. This is to allow the trailer to track back in line before entering the second curve. Should curves need to be closer than 140FT apart the road width will need to be increased to give the trailer adequate room to follow around the corner.

This can be looked at on case-by-case basis.

Grades

The maximum grade a road truck can safely traverse depends greatly on the road surface.

Compaction, gravel type and size, and weather all play critical roll in a road truck being able to pull a load safely up or down a grade.

As a rule any non-paved grade exceeding 5% may need an assist vehicle in adverse weather conditions, or if road surface lacks the necessary level of compaction to maintain traction.

Any site with grades ranging from 5% to 7% non-paved will need to have an assist vehicle on stand by during adverse weather or road conditions.

Within a 7%+climb, there can be no more than a 30-degree curve at any point and no more than 1 curve within 100 meters.

Any site with grades exceeding 7% non-paved will need to have an assist vehicle on site at all times.

Down grades exceeding 7% can be looked at on a case-by-case basis to determine if an assist vehicle will be needed to assist in braking. This will be based on road surface, grade percentage, length of grade, and weather conditions.

10%+

Any grades over 10% will need to be looked at on case-by-case basis.

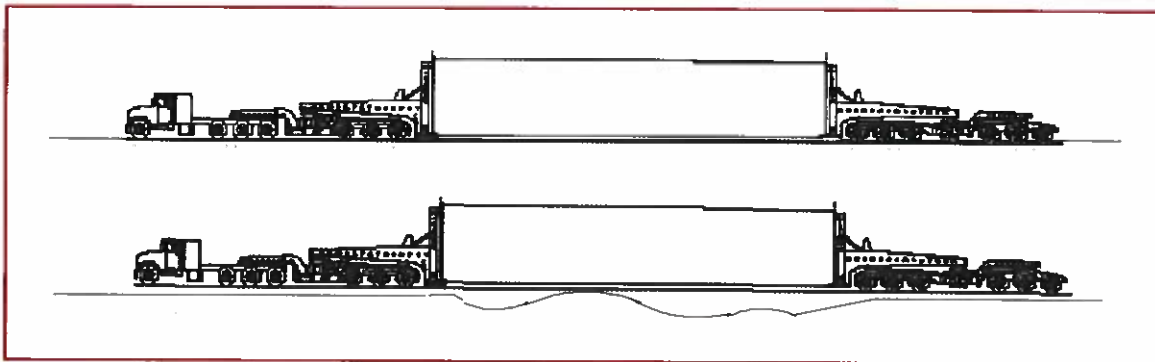
Unloading Areas

The area around the foundation and crane pad that trucks must maneuver in shall be cleared, leveled and compacted in such a way as to support the weight of the loaded trucks as well as the components when staged.

Trucks range from 80FT long to 180FT long and have a gross weight of up to 260,000lbs.

The unloading area will need to be level with the road and crane pad. This will allow the truck to drive off the road into the unloading area.

The unloading area shall be as flat as possible for the length of the truck and trailer.



Uneven unloading area will make positioning the tower in the needed location difficult. It will also be difficult to block and unload the tower section if on a Schnable type trailer.

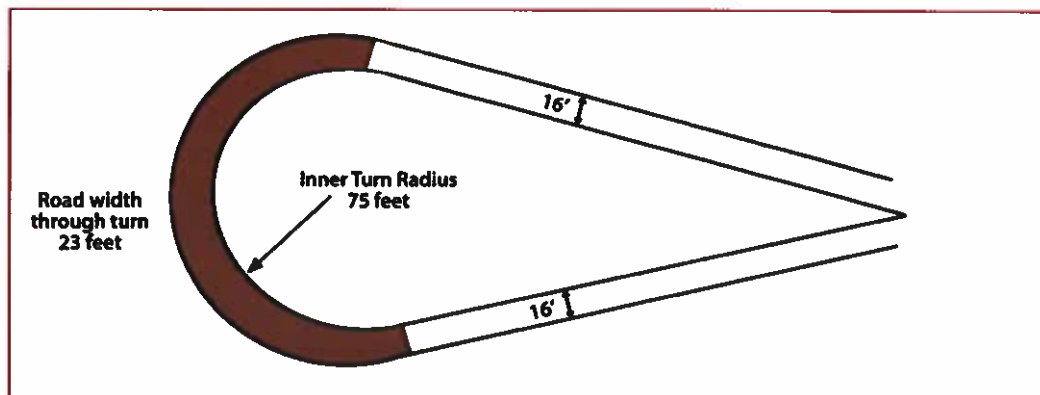
The unloading area shall be laid out in such a way that trucks can, drive in loaded, unload, and then turn round empty. See: Pad lay out example.

Should trucks need to turn around in order to unload, a turn round area shall be built that is capable of supporting the loaded truck, adhering to all road requirements.

Unloading Areas

The end of all string roads shall have a way to turn around empty trucks.

To turn around an empty 13-axle non-steer trailer a loop or jug handle type turn that has a 75Ft inner radius and a 23Ft wide road surface through the turn.



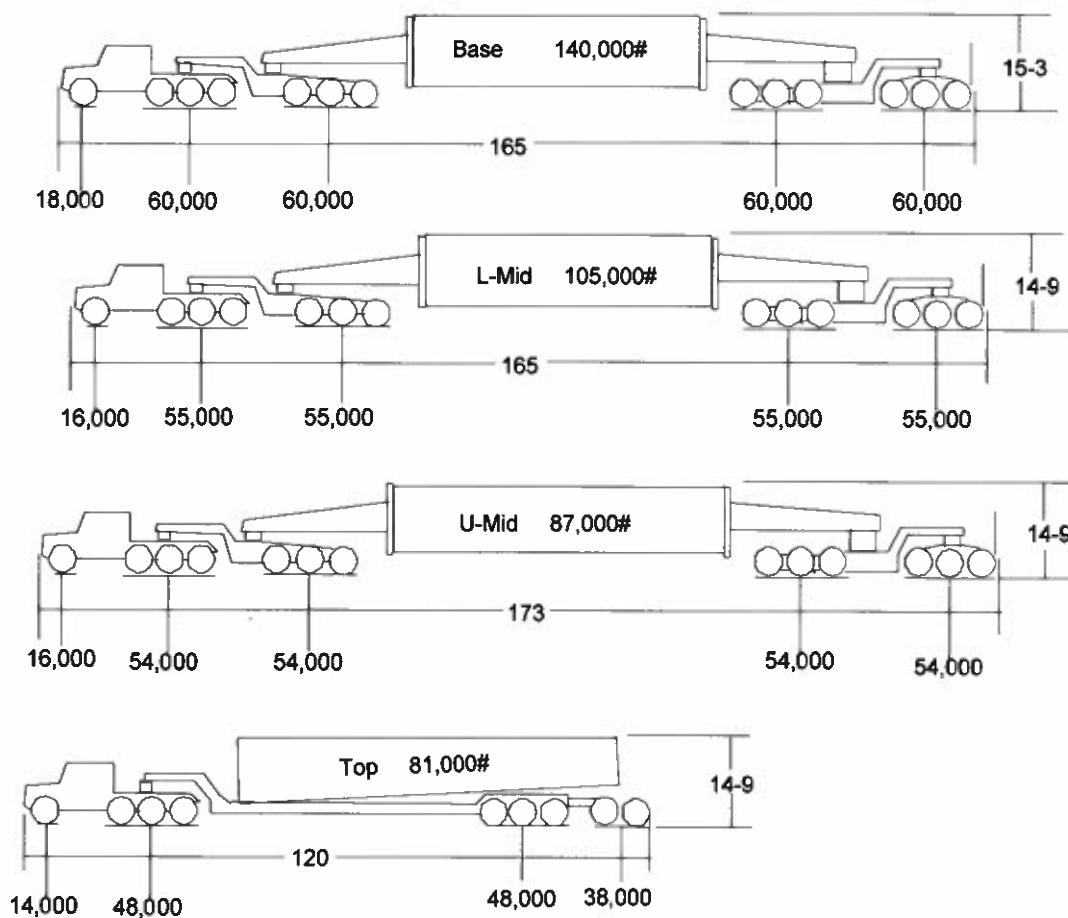
All empty trucks could use this type of turn round.

The maximum empty gross weight of the multi axle trailers is 98,000lbs. Any turn round will need to be able to support this weight.

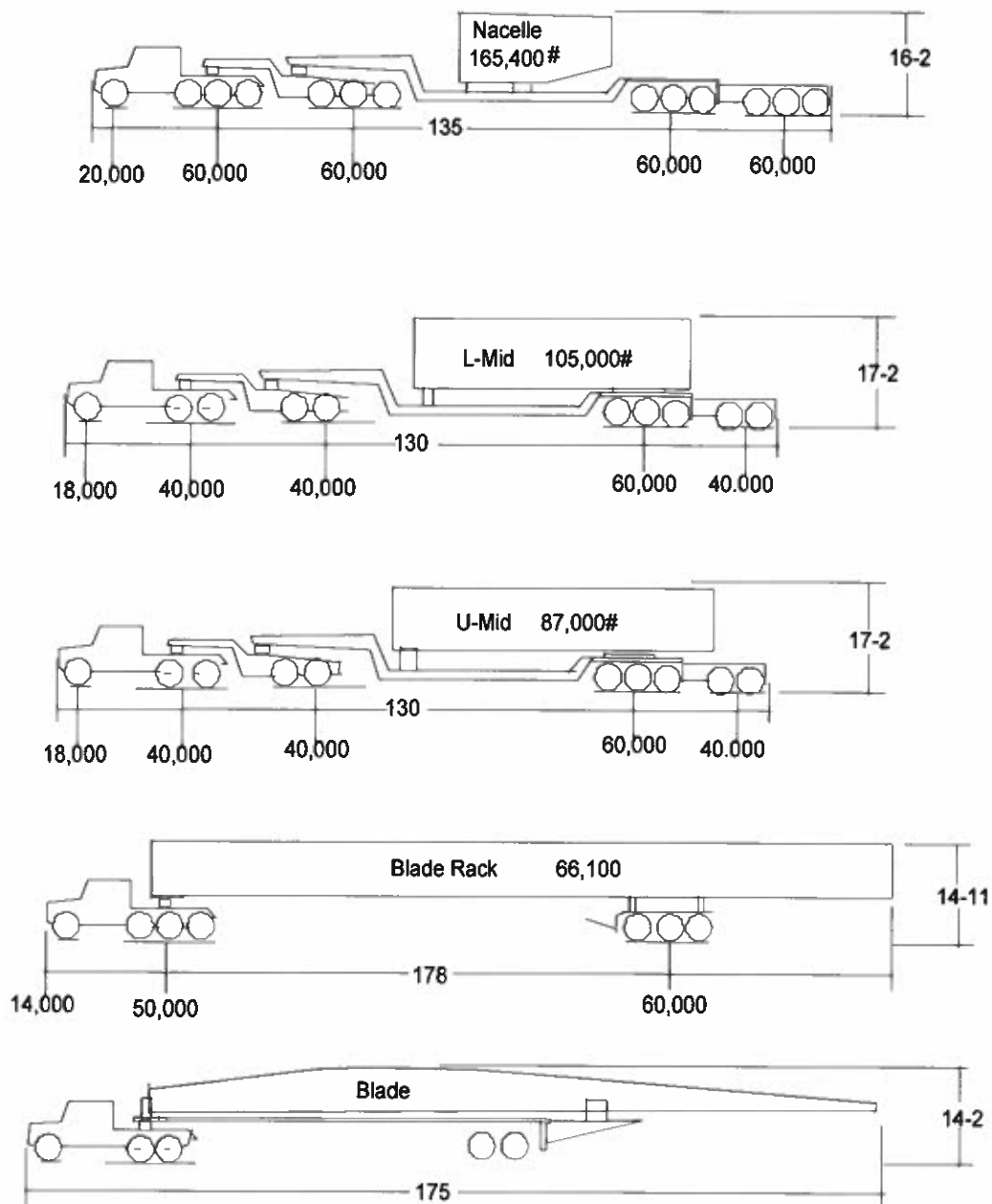
To turn loaded trucks around the turnaround areas would need to be about twice the size. The inner radius would need to be 135Ft with a 23Ft wide road surface. This would also have to be built to support the weight of loaded trucks.

Weights, Length and Height

of Suzlon S-88 components loaded on various types of truck and trailers. The diagrams below show Maximum dimensions and weights of transport vehicles.



Weights, Length and Height (continued)



Suggested Pad Layout

This is a layout for pads located adjacent to the road. The trucks would be able to drive forward onto the pad eliminating the need to be turned around loaded. After unloading the trucks would be able to travel to the last pad on a string to turn around.

This pad would be able to be mirrored for foundations located on the opposite side of the road.

This pad could be used at the end of a string as long as the road ends adjacent to the last foundation and the crane pad is put in at an angle, in the same manner as the pads along the side of a string road. The turn-a-round could then be continued from the point the road ends, in the form of a loop or jug handle. This would allow unladen trucks a place to turn around at the end of a string.

This layout dose not show a location for the Blades due to the fact some Blades come in racks and some come as singles. This can be modified for specific site requirements.

Using this layout would require the Base, U-Mid, and L-Mid Tower sections to be loaded with the "TOP" end of the tower to the front end of the tractor/trailer. (On both Schnable trailers and on Deck trailers) The Top Tower section would be loaded with the "TOP" end of the tower section to the rear of the tractor/trailer. (This is the only way a top can be loaded.)

The reason for this type of pad layout is;

1. All trucks would be able to drive forward into a pad, eliminating the need to try and turn loaded trucks around or back them up.

Suggested Pad Layout (continued)

2. In the event inclement weather causes bad site conditions, resulting in the truck needing to be towed. Trucks could be pulled forward, directly into location. This would almost eliminate the need to pull the truck backwards.
3. By the trucks being able to pull forward into a pad it will reduce unloading times, by eliminating the time needed to back or jockey around to get into position.

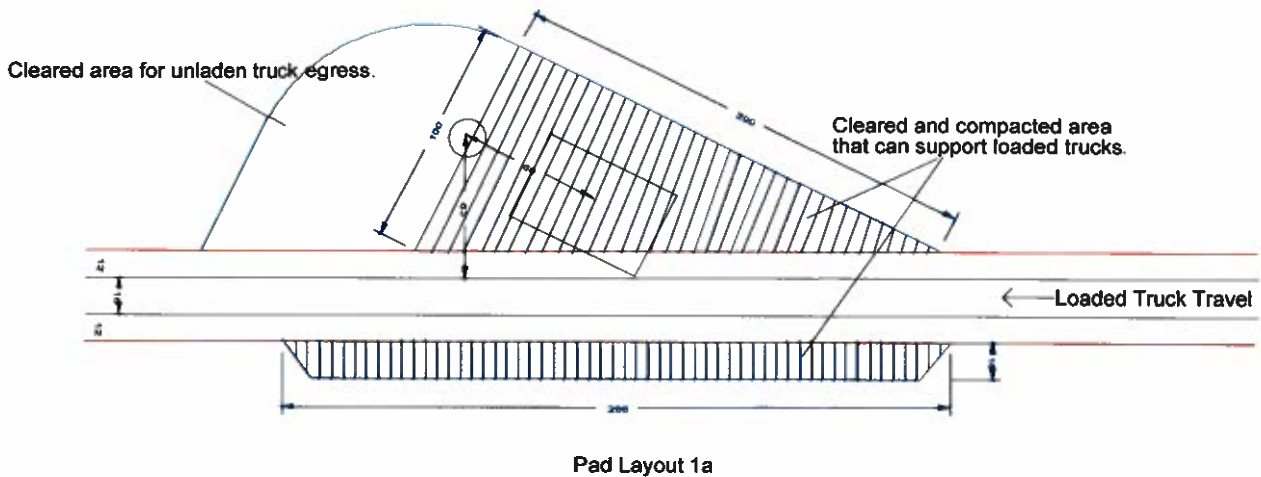
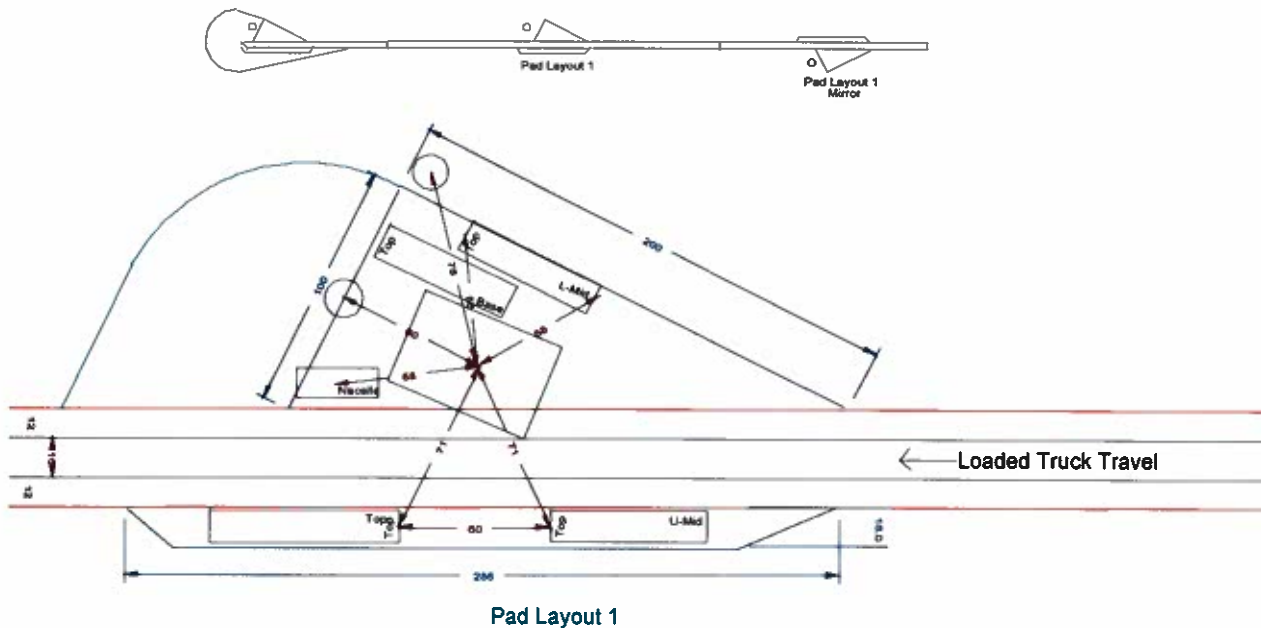
The area that unladen trucks use to exit the pad needs to be cleared and capable of supporting an unladen truck.

Pad layout 1 diagram; this example shows placement, position and orientation of components staged at a WTG pad. This example diagram shows a 60x40 crane pad and shows component locations based on a crane with a minimum-lifting radius of 55 feet. . It also shows the Base tower section placed so that a smaller crane can set the base before the main erection crane arrives at the pad. The pads are also configured so the main erection crane only needs to turn at 45-degree angles to walk onto the pad.

***The exact size of the crane pad and the exact distance of each component from the center pin of the crane would be determined by the erection company based on their needs and the type of equipment they are using to erect the WTG. ***

If the Base and L-Mid need to be placed farther away, the width of the lay-down area may need to be widened.

Suggested Pad Layout (continued)



ATTACHMENT E

ATTACHMENT E. COST ESTIMATE



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
1	Road Design	Professional Level 10	\$176	16	\$2,616.00
		Professional Level 7	\$131	56	\$7,336.00
		Technical Level 3	\$72	8	\$576.00
		Administrative 3	\$72	2	\$144.00
	Activities				
	- Project management and coordination				
	- Evaluate route alternatives				
	- Prepare design drawings				
	- Coordinate construction cost estimate with DHP				
	- Prepare project memo and submit to the City				
	Assumptions				
	- One round of Duluth/City comments will be incorporated into the final design				
	- Final comments will note require extensive redesign				
	Labor				\$10,672.00
	Equipment			0	\$0.00
	Travel			0	\$0.00
	Per Diem			0	\$0.00
	Miscellaneous			0	\$0.00
	Expenses				\$0.00
	Other			0	\$0.00
	Subcontractors				\$0.00
	Activity 1 Subtotal				\$10,672.00

ATTACHMENT E. COST ESTIMATE



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
2	Construction Surveying	Professional Level 10	\$176	2	\$352.00
		Professional Level 7	\$131	6	\$786.00
		Professional Level 5	\$104	24	\$2,496.00
		Professional Level 2	\$73	16	\$1,168.00
		Administrative 3	\$72	2	\$144.00
					<u>\$0.00</u>
	Activities				
	- Project management and coordination				
	- Stake centerline alignment at horizontal and vertical alignment points and 100' intervals thereafter				
	- Perform as-built survey upon completion of construction				
	Assumptions				
	- Staking will be performed over the course of two days				
	- Survey crew will mobilize from Laramie, Wyoming				
	- As-built survey will be performed in one day				
		Labor			<u>\$4,946.00</u>
		GPS-Survey Grade	\$420	3	\$1,260.00
		Company Vehicle	\$88	3	\$258.00
		Per Diem	\$45	0	\$0.00
		Supplies	\$250	1	<u>\$250.00</u>
		Expenses			<u>\$1,768.00</u>
		Other		0	<u>\$0.00</u>
		Subcontractors			<u>\$0.00</u>
		Activity 2 Subtotal			<u>\$6,714.00</u>

ATTACHMENT E. COST ESTIMATE



Activity		Schedule of Charges	Hourly Rate or Charge	Hours or Unit Estimate	Subtotal
3	Construction Oversight	Professional Level 10	\$178	4	\$704.00
		Professional Level 7	\$131	8	\$1,048.00
		Professional Level 4	\$94	15	\$1,410.00
		Technical Level 4	\$62	45	\$3,690.00
		Technical Level 3	\$72	10	\$720.00
		Administrative 3	\$72	2	\$144.00
					<u>\$0.00</u>
	Activities				
	- Project management and coordination				
	- Observe construction activities and verify work is completed in accordance with the Construction Drawings				
	- Perform subgrade, embankment, and surfacing density testing at specified frequencies				
	- Prepare daily project summaries				
	- Prepare as-constructed drawings				
	Assumptions				
	- Construction will occur over three weeks				
	- Project summaries will be incorporated into the HJLF Cells 1, 2, and 3 Excavation Project				
	- Construction oversight will be conducted in conjunction with HJLF Cells 1, 2, and 3 Excavation project				
	- Materials testing assumptions				
	- 4 proctors will be needed				
	- 4 Atterberg limits				
	- 4 Gradations				
	- Nuclear density gauge charged over 3 weeks which includes conducting density tests				
		Labor			<u>\$7,718.00</u>
		Soils Testing	\$2,400	1	\$2,400.00
		Mileage	\$0.58	150	\$84.00
		Per Diem		0	\$0.00
		Miscellaneous		0	<u>\$0.00</u>
		Expenses			<u>\$2,484.00</u>
		Other		0	<u>\$0.00</u>
		Subcontractors			<u>\$0.00</u>
		Activity 3 Subtotal			<u>\$10,200.00</u>

	Trihydro Total	Add Client Info	TOTAL
Task 1 Road Design	\$10,872	\$0	\$10,872
Task 2 Construction Surveying	\$8,714	\$0	\$8,714
Task 3 Construction Oversight	\$10,200	\$0	\$10,200
Total	\$27,786	\$0	\$27,786